TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Preparation of Different Types of Standard Solution

Primary and secondary standards, molarity and normality, and the weighing-and-dilution procedure by which an exactly known solution is made.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 11 · LEVEL 1 TERM 1

1. Aim and definitions

Aim: to prepare standard solutions of known concentration accurately. A **standard solution** is one whose concentration is known exactly; every titration in Experiments 13–15 hangs on one. Concentrations are expressed as:

- **Molarity (M)** — moles of solute per litre of solution.
- **Normality (N)** — gram-equivalents per litre; equivalents count reactive units (for acids, replaceable H^+; for bases, OH^- accepted), so N = M × (basicity or acidity). Equal volumes of equal-normal solutions neutralise exactly — the convenience that built classical volumetric analysis.

$$\text{mass needed (g)} = \text{M (mol L}^{-1}) \times \text{volume (L)} \times \text{molar mass (g mol}^{-1})$$

2. Primary and secondary standards

KIND	CHARACTER AND EXAMPLES
Primary standard	A substance pure enough, stable enough and unchanging enough to be weighed directly into a standard solution: definite composition, high purity, not hygroscopic or efflorescent, unchanged in air, high molar mass (to shrink weighing error), readily soluble, reacting stoichiometrically. Examples: anhydrous Na_2CO_3, oxalic acid dihydrate ($H_2C_2O_4 \cdot 2H_2O$), $K_2Cr_2O_7$, NaCl.
Secondary standard	A substance failing one or more of those tests — NaOH (absorbs water and CO_2), HCl (a gas in solution, concentration drifts), $KMnO_4$ (decomposes slowly). Its solutions are made approximately and then **standardised** against a primary standard — Experiments 13–15 are exactly that operation.

3. Procedure: making 250 mL of 0.1 M Na_2CO_3 (primary standard)

1. **Calculate:** moles = 0.1 × 0.250 = 0.025 mol; mass = 0.025 × 106 = **2.65 g** of anhydrous Na_2CO_3.
2. **Weigh accurately** (analytical balance, by difference from a weighing bottle) a quantity close to 2.65 g; record the exact mass — the solution's true molarity is computed from the actual mass, not the target.
3. **Transfer quantitatively** through a funnel into a 250 mL volumetric flask; rinse bottle and funnel into the flask (the rinsings are part of the transfer — a gram left behind is a wrong concentration).
4. **Dissolve** in about half a flask of distilled water, swirling.
5. **Make up to the mark:** add water until the meniscus's bottom (read at eye level) sits exactly on the etched line; stopper and invert repeatedly to homogenise.
6. **Label:** substance, nominal and exact concentration, date, preparer.

WORKED EXAMPLES — QUANTITIES FOR COMMON PREPARATIONS

(a) 0.1 M oxalic acid, 250 mL: $H_2C_2O_4 \cdot 2H_2O$ has molar mass 126 g mol^{-1}; mass = 0.1 × 0.250 × 126 = **3.15 g**. **(b)** 0.1 N oxalic acid: oxalic acid is dibasic, equivalent mass = 126/2 = 63; for 250 mL, mass = 0.1 × 0.250 × 63 = **1.575 g** — half the molar preparation, as it must be: 0.1 N = 0.05 M for a dibasic acid. **(c)** If the actual mass weighed in the Na_2CO_3 preparation was 2.63 g, the exact concentration is (2.63/106)/0.250 = **0.0992 M** — and that, not "0.1 M", is what Experiments 13–15 must use.

4. Dilution of a stock solution

To prepare a dilute standard from a concentrated one, moles are conserved: $\mathbf{M_1V_1 = M_2V_2}$. The concentrated volume is measured by pipette or burette (never by mouth pipetting — a pipette filler is used), delivered into the volumetric flask, and made up to the mark. Experiment 12 applies this to the two great bench acids.

5. Precautions

- Use distilled water only; rinse glassware with the solution it will hold where the procedure directs (burette and pipette), but the volumetric flask need only be clean, not dry — the solute is made up to volume regardless.
- Read the meniscus at eye level, bottom of meniscus on the line for colourless solutions.
- Weigh hygroscopic or hot materials only after cooling in a desiccator; never weigh directly on the balance pan.
- A standard solution is labelled the moment it is made — Experiments 13–15 grade the label as well as the liquid.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. List four properties of a primary standard and name two examples; then state why NaOH fails the test.
2. Calculate the mass of $K_2Cr_2O_7$ (M = 294 g mol^{-1}) needed for 500 mL of 0.05 M solution.
3. Convert: a 0.25 M H_2SO_4 solution is what normality? A 0.1 N NaOH solution is what molarity?
4. In the worked example, why is the solution's concentration computed from 2.63 g rather than reported as 0.1 M?
5. 50 mL of 1.0 M HCl is diluted to 500 mL. Find the new molarity using $M_1V_1 = M_2V_2$.
6. Why is the volumetric flask not required to be dry inside, while the weighing bottle must be?

Answer key

1. Properties (any four): high purity; definite, unchanging composition; stability in air (not hygroscopic/efflorescent); high molar mass; ready solubility; stoichiometric reaction. Examples: Na_2CO_3, oxalic acid dihydrate (also $K_2Cr_2O_7$, NaCl). NaOH fails: it absorbs moisture and CO_2 from air, so a weighed mass is not a known amount of NaOH — its solutions must be standardised.
2. Mass = 0.05 × 0.500 × 294 = 7.35 g.
3. H_2SO_4 is dibasic: 0.25 M = 0.50 N. NaOH is monoacidic base: 0.1 N = 0.1 M.
4. Because a standard solution's authority is its actual concentration: the weighing delivered 2.63 g, not the nominal 2.65 g, and every later calculation inherits the error if the nominal value is used. Exactness lives in using the measured quantity.
5. $M_2 = M_1V_1/V_2 = 1.0 \times 50/500 = 0.10$ M.
6. The flask is filled to a fixed volume mark — residual rinse water becomes part of the measured volume and changes nothing. The weighing bottle's contents are measured by mass; moisture in it would be weighed as if it were solute, corrupting the mass itself.